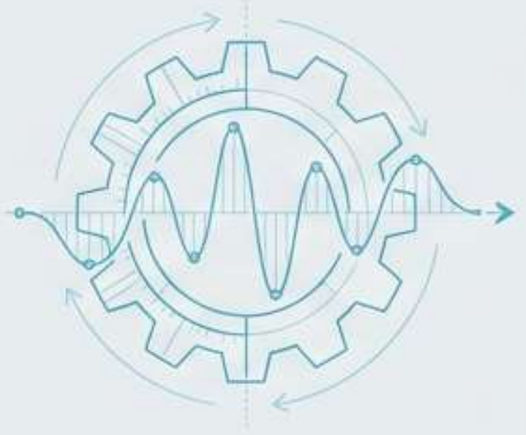
AI Predictive Maintenance

Machine Failure Prediction

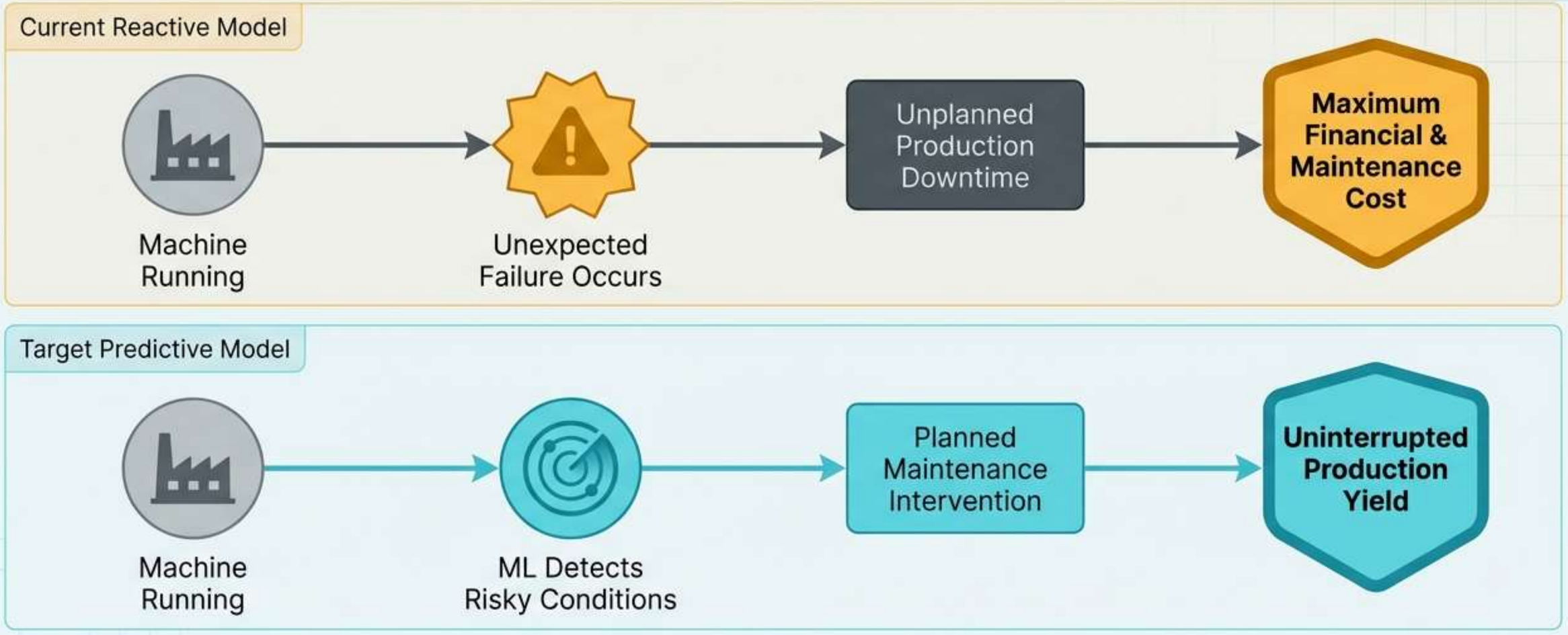
Moving factory operations from reactive repairs to proactive prevention.



By leveraging machine learning on continuous sensor telemetry, we can identify and resolve hidden mechanical degradation before costly failure events occur.

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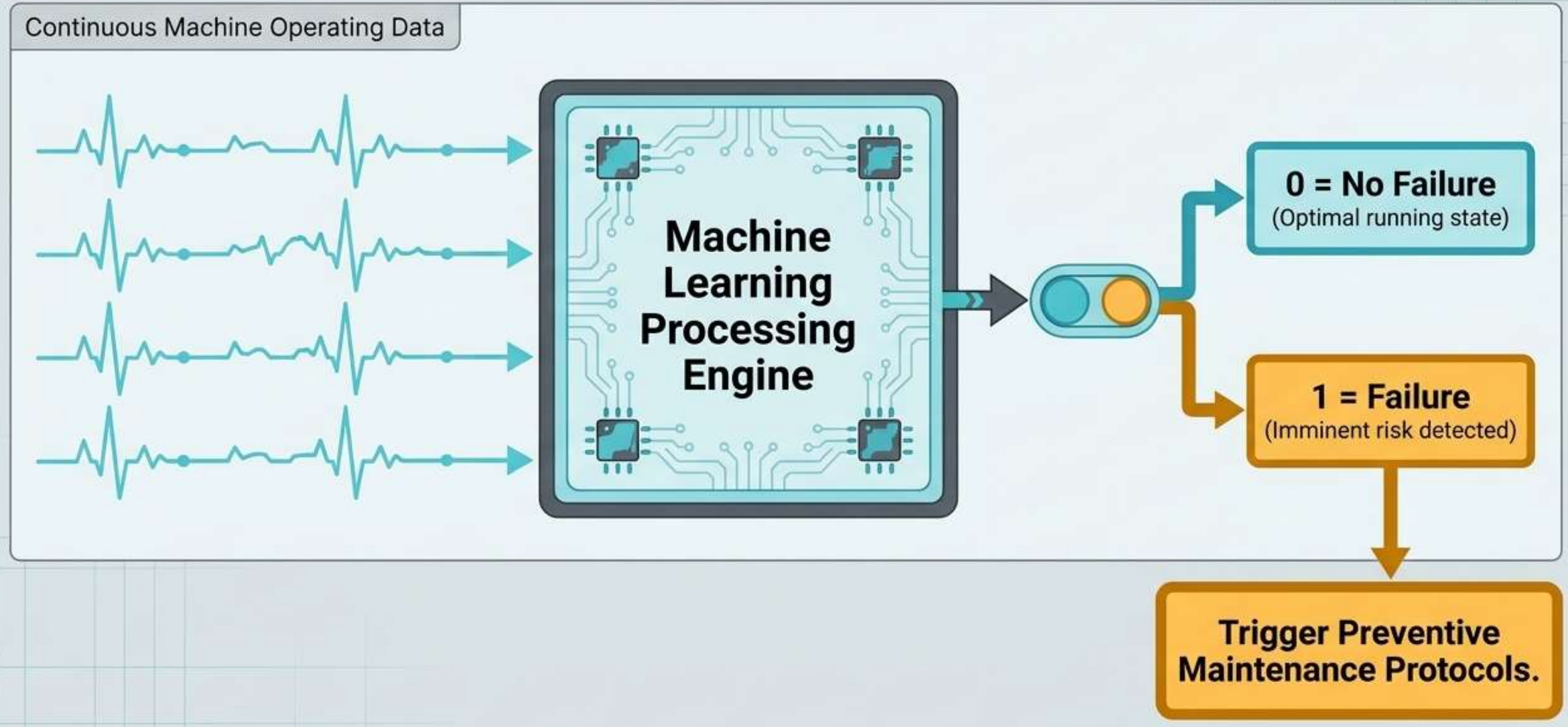
The Cost of Unexpected Downtime



The operational goal is to transition entirely to the predictive track—identifying anomalous physical conditions in the data stream before the machine reaches a failure state.

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The Machine Learning Intervention Engine



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The AI4I Training Environment

AI4I 2020 Predictive Maintenance Dataset



Source: UCI Machine Learning Repository
(RWTH Aachen University)

10,000
Total Records

14
Feature Columns



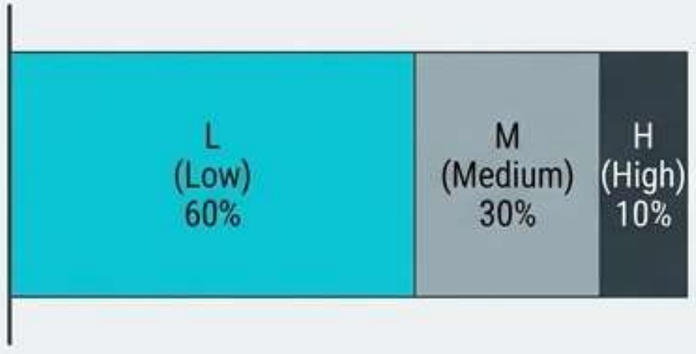
Synthetic Industrial Telemetry

Carefully designed to **simulate highly realistic machine operating conditions** and physical sensor relationships **without live factory privacy constraints.**

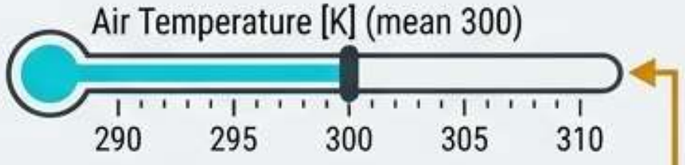
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Mapping the Telemetry Sensors

Product Quality (Categorical)



Thermal Sensors

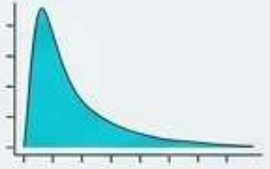


High Correlation Detected ($r \approx 0.88$).
Process temperature is driven by ambient air.
Risk of multicollinearity in linear models.



Mechanical Sensors

Rotational Speed [rpm]



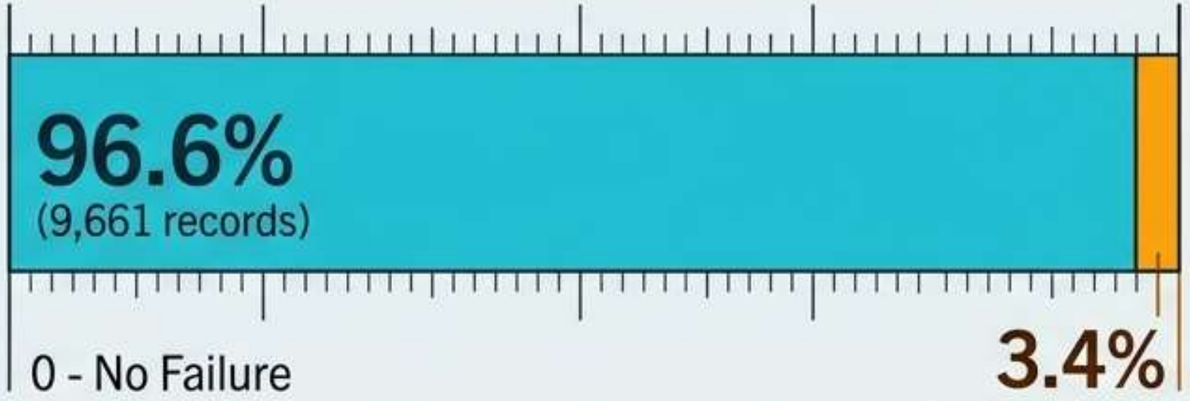
Torque [Nm]

Tool Wear [min]
Ranges evenly 0-253, resets upon maintenance

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System Constraints and Leakage Hazards

Target Variable: Machine Failure



Extreme Class Imbalance. Accuracy alone will be misleading; requires precision/recall/F1 scoring and resampling (e.g., SMOTE).

Data Leakage Risks & Exclusions

Administrative Drops

Exclude UDI & Product ID (Identifiers with no predictive value).

CRITICAL: Target Sub-Flags

TWF | HDF | PWF | OSF | RNF

⚠ **Do not feed these sub-flags as input features.** They directly leak the target answer and will invalidate the predictive model.

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